What Are the Capabilities of Image Analysis?

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How Do We Use Vision?

As a human we use vision in several ways.

- Identify what we are looking at
- Describing what we see
- Finding objects in a picture
- Reading text
- Identify people
- Describe a video

Sometimes we have varying levels of confidence!

Artificial Intelligence vision services provide computer-enabled similar capabilities

Image Analysis API Versions

3.2

Tags, Objects, Descriptions, Brands, Faces, Image type, Color scheme, Landmarks, Celebrities, Adult content, Smart crop

4.0

Read text, Captions, Dense captions, Tags, Object detection, Custom image classification / object detection, People, Smart crop

Computer Vision – Image Analysis

Model customization (4.0)	People detection (4.0)	OCR (4.0)
Caption and tag generation	Object detection	Smart cropping/area of interest

Computer Vision – Image Analysis

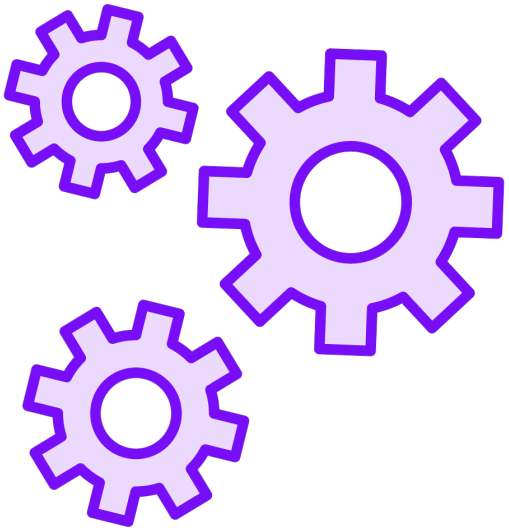
Background removal (4.0)

Multimodal embeddings (4.0)

Product Recognition/detect brands

Supporting Azure AI Resource

Azure AI Vision



Multi-service

Demo: Analyzing an Image

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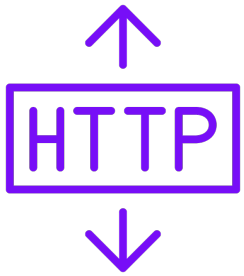
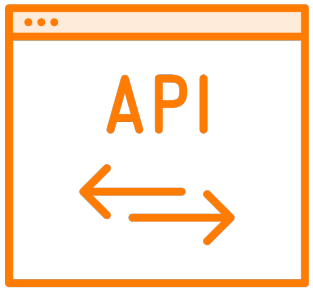
Ways to Analyze an Image

Vision Studio



REST API



SDK

Analyze Parameters

Features

[Model-name]

[Language]

[Smartcrops-aspect-ratio]

[Gender-neutral-caption]

Using REST

PowerShell Example

```
$endpoint = $env:AI_SVC_ENDPOINT
$read_image_url = “image.jpg"

$ai_url = $endpoint +
"computervision/imageanalysis:analyze?api-version=2024-02-
01&features=read,objects,tags&language=en"

$headers = @{
    'Content-Type'='application/json'
    'Ocp-Apim-Subscription-Key'=$env:AI_SVC_KEY
}
$body = @{
    "url"=$read_image_url
} | ConvertTo-Json

$resp = Invoke-RestMethod -Uri $ai_url -Method Post -
Headers $headers -Body $body
```

Using SDK

Python

```python
from azure.ai.vision.imageanalysis import
ImageAnalysisClient
from azure.ai.vision.imageanalysis.models import
VisualFeatures
from azure.core.credentials import AzureKeyCredential

features = [VisualFeatures.TAGS, VisualFeatures.OBJECTS,
VisualFeatures.READ]

# Analyze the image
result =
imageanalysis_client.analyze_from_url(image_url=url,
visual_features=features)
```

Generating a Thumbnail

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Why We Want Thumbnails!

Very often we will have an image that we need a version of in a different aspect ratio.

We don't want to skew the original image as this would result in a distorted result.

The main subject may be only in a portion of the image, which should be the central point of a new image.

Thumbnails provide an area of the picture at the desired aspect ratio, containing the primary subject.

While face detection is used to detect the presence of a person, there is no distinguishing of specific faces.

Supported Aspect Ratios

Aspect ratio is the relationship between the width and height of an image.

A ratio of 3:2 would be 1.5.

Valid aspect ratios are between .75 and 1.80.

1

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1.5

Using Smart Crop

REST Example

```
ai_url = endpoint +
"computervision/imageanalysis:analyze?api-version=2024-02-
01&features=SmartCrops&smartcrops-aspect-ratios=1,1.5"

...

resp.smartCropsResult.values
```

Using Optical Character Recognition (OCR)

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What Is OCR?

Optical Character Recognition (OCR) is the ability for computers to identify characters from an image.

When characters are placed together, they form words, sentences and complete documents.

Very often there are scenarios where extracting the text from an image is very useful.

- Accessibility
- Automation
- Form and document processing

AI Solutions for OCR

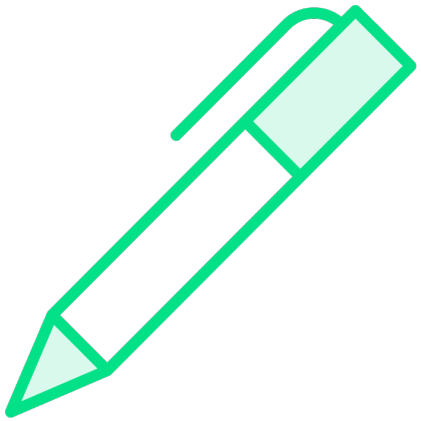
Read with Image Analysis

Ideal for small amounts of text.
Works with images only.
Synchronous API.

Document Intelligence

Used for larger amounts of text.
Supports document formats.
Asynchronous API.

Handwritten vs. Print Text



The Read in Image Analysis v4.0 supports both handwritten and print text.

The languages supported differ with print text having greater language support.

Different languages within the same image are supported providing you do not specify a language code.

Image Analysis Review

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Summary

There are many image analysis capabilities including:

- Object and people detection
- OCR
- Caption and tagging
- Smart cropping

Version 3.2 and 4.0 of API available.

When calling the API there are various parameters.

Can use REST and the SDK.